

Chest X-ray Pneumonia Detection

. Introduction

I chose the Chest X-ray Pneumonia dataset because medical image analysis is a critical application of deep learning that can assist doctors in making faster and more accurate diagnoses. The goal is to build a system that can automatically distinguish between normal lungs and lungs affected by pneumonia from X-ray images.

. Dataset Description

The dataset contains X-ray images. For the training phase, I used images categorized into two classes: 'NORMAL' and 'PNEUMONIA'.

- NORMAL: images
- PNEUMONIA: images Since the classes are imbalanced (Pneumonia cases are significantly higher), the F-score was chosen as the primary metric for evaluation to ensure the model doesn't just predict the majority class.

. Models Used

I trained three traditional machine learning models using flattened pixel data (x grayscale images):

- . Logistic Regression: Achieved an F-score of It performed surprisingly well, finding a strong linear separation between classes.
- . Random Forest: Achieved an F-score of It was very good at recall but slightly less precise than Logistic Regression.
- . KNN (K-Nearest Neighbors): Achieved an F-score of It was the weakest among the three, as it struggled with the high dimensionality of raw pixel data.

. Neural Network

I built a Deep Neural Network (ANN) using Keras. The architecture included:

- An Input layer for the , flattened pixels.
- Two Dense layers (and neurons) with ReLU activation.

- A Dropout layer (.) to prevent overfitting by randomly disabling neurons during training.
- An Output layer with a Sigmoid activation for binary classification. I also used Early Stopping to stop training when the validation loss stopped improving.

. Results & Comparison

The Neural Network achieved an F-score of which is better than KNN and similar to Random Forest. However, Logistic Regression remained the top performer in this specific flattened-pixel scenario. While the Neural Network is more complex, it showed great stability and low overfitting due to the Dropout layer. For this size of dataset and flattened input, the extra complexity of the ANN didn't significantly outperform the linear efficiency of Logistic Regression.

. What I Learned

I learned that more complex models are not always better for every type of data representation. While Neural Networks are powerful, traditional models like Logistic Regression can be very effective if the data is prepared correctly. I also learned the importance of Dropout and Early Stopping in preventing a model from simply memorizing the training data, ensuring it performs well on new, unseen images.

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